

1 · The telemetry contract test. Run it against your own stack.

A dashboard depends on metric names and attributes the same way code depends on an API. Write that dependency down and test it. The test drives real traffic and checks five things per instrument: **C1** emitted at runtime · **C2** required attributes present · **C3** convention stability · **C4** SDK helper carries the documented bucket advice · **C5** the cross-team join is possible. **My own dashboard passed 4 of its 11 assumptions.** The three failures worth copying:

check	what it caught (grpc-python 1.83 / semconv 0.65b0)
C1	grpc.client.call.duration is documented as "stable, on by default" and is never emitted by the Python build
C4	create_gen_ai_server_time_to_first_token() passes no bucket advisory → SDK's 15 defaults (top bucket 10,000 ms) instead of the 14 documented boundaries → a true ~14 ms p50 reports as ~2,500 ms
C5	gen_ai.server.* attrs {operation.name, provider.name, request.model} n grpc.* attrs {method, target, status} = EMPTY SET . Nothing to group by jointly



2 · The tail-sampling inversion. The evidence behind the one action.

stream dur	dur ÷ decision_wait	seen by sampler	kept (want: slow)
0.76 s	0.38	758 ms	0/6 (fast, correct)
1.28 s	0.64	1,277 ms	6/6 (slow, correct)
3.2 s	1.61	51 ms	0/6
6.4 s	3.22	51 ms	0/6
12.7 s	6.33	51 ms (245× under)	0/6

A span reaches the sampler only at `Span.End()`; the latency policy takes `max(end)–min(start)` over the spans it holds. Past `duration = decision_wait` (default **30 s**) it judges your stream by the spans that finished early. Detection **inverts**: slower requests become less likely to be kept, every time. Turning on the decision caches converts partial recovery into consistent total loss. So the histogram has to tell the story on its own. Two histograms cost about 2.2 KB per series per interval, bounded by buckets, not traffic.

The one action: compare your sampler's `decision_wait` to your p99 stream duration; then `OTEL_METRICS_EXEMPLAR_FILTER=always_on`

Measured on a re-implementation of the processor that follows its README line by line (each behaviour is quoted beside the code), not on the Collector binary. Reproduce it on yours. I want to know if it doesn't.

3 · The naming gap. One signal, three spellings, no join key.

owner	the same signal is spelled	legal OTel instrument?	slice-by attributes
model server (shipped)	vllm:time_to_first_token_seconds	NO . ':' is outside the name grammar	model, engine labels
GenAI semconv (Development)	gen_ai.server.time_to_first_token	yes	gen_ai.operation.name, gen_ai.provider.name
gRPC A66 (today)	grpc.client.attempt.duration <i>only</i>	yes	grpc.method, grpc.target, grpc.status
this talk's ask	grpc.client.attempt.time_to_first_message	yes	A66's, verbatim

7 of 14 shipped metric names across vLLM and SGLang cannot be registered as OTel instruments. The Collector's Prometheus receiver renames them, so nobody is blocked. But the rename carries no record of where it came from (gRPC's own plugin sets no `schema_url`), so a renamed series and a dead series look identical on a panel: "No data". And the two teams' attribute sets share **no dimension**. The translation step between them is real, owned by nobody, and invisible on both dashboards.

The ask: two histograms, **time to first message** and **inter-message gap**, at the transport read boundary, client side first, with A66's buckets and attributes, experimental and off by default per A79's stability guidelines. It is the shape A96 shipped (merged 2025-07) and A108 templates. Necessary. And honestly, not sufficient: it does not create the join key.

Sources: A66 · A79 · A96 · A108 · A80 · GenAI metrics (semconv-genai repo) · tail sampling processor · vLLM v1 metrics design · OTel versioning & stability · grpc.io OTel metrics guide

All measurements are loopback, single host, CPython, on a simulated cost model. Python-build findings are labelled as such. Go and Java builds are unchecked. GenAI semconv is Development status: breaking changes may occur, and its docs moved repos in 2026. The links above point at the living copy.